AthenaStudies

Intelligent Systems: Fuzzy Logic

Explanation Slides | Mara Spadon | 2023

Vagueness and Uncertainty

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Vagueness: Lack of clarity or precision which leaves room for interpretation.

Uncertainty: The information provided leaves doubt about its accuracy or completeness. It implies that there is something not fully known within the context of the statement.

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Vagueness: Lack of clarity or precision which leaves room for interpretation.

"The weather is warm today."

In this statement, the term "warm" is subjective and lacks precision. Different people may interpret "warm" differently based on their preferences and experiences. What is considered warm to one person might be cool to another. The lack of a specific temperature range makes the statement **vague**, leaving room for interpretation.

Uncertainty: The information provided leaves doubt about its accuracy or completeness. It implies that there is something not fully known within the context of the statement.

"A significant portion of the company employees participated in the training program."

In this statement, the use of the term "significant portion" implies a considerable number of employees, but the exact figure is unknown. While it suggests a substantial participation rate, the lack of specific details regarding the actual number or percentage of employees who attended the training introduces **uncertainty**. Without supporting information or concrete data, the accuracy of the claim about the extent of employee participation remains **uncertain**.

Uncertainty and Vagueness: Practice Question

Exercise 1: Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A) The author of this statement describes the marital status of Dutch men with a degree of vagueness.
B) The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

Give an example for a statement for the other property?

Uncertainty and Vagueness: Practice Question

Exercise 1: Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A) The author of this statement describes the marital status of Dutch men with a degree of vagueness.

B) The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

Give an example for a statement for the other property?

This statement is described with a **degree of uncertainty**.

Upon hearing the statement "Half of the Dutch men are married," it gives the impression of a precise figure, yet its accuracy remains uncertain without any supporting information.

While it is presented as an exact half, the true percentage could be different. Essentially, the accuracy of the claim is unknown due to the lack of specific details for verification.

Vagueness and Uncertainty: Example Question

Exercise 1: Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A) **The author of this statement describes the marital status of Dutch men with a degree of vagueness.**
B) The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

Give an example for a statement for the other property?

If a statement would be described with a degree of vagueness, we could consider the following answer: "**Dutch men are very tall.**"

The expression "very tall" is subjective and its interpretation may differ depending on the individual's perspective.

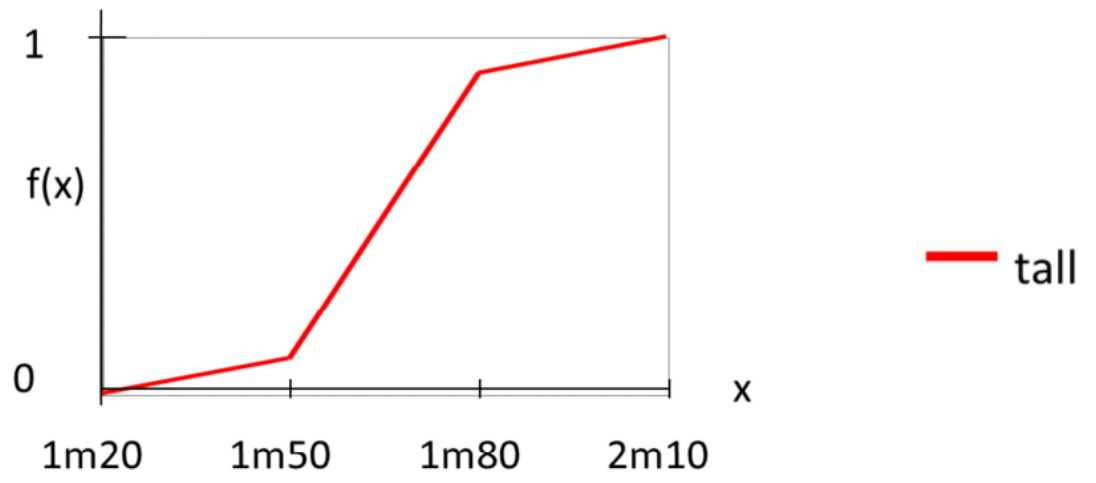
Fuzzy Set Theory and Membership

Fuzzy Set Theory and Membership

Fuzzy set theory: Elements can belong to a set to a certain degree.

Membership function: Returns a number between 0 and 1 where 0 means no membership and 1 means full membership.

$f_A(x) = 1$: x is a member of A *(certainly)*.
$f_A(x) = 0$: x *not* a member of A *(certainly)*.



Graph of the membership function of the fuzzy set of tall people.

Fuzzy Set Theory: Recap Question

Exercise 2: The director of our institute wants to reduce the environmental impact of the company by reducing the CO2 emissions caused by transportation. Therefore, she wants to give a devise a policy to support those who do not live very close (they can come by bicycle) or those who live very far (they should do remote work).

Since you know the director is a big fan of fuzzy set theory, you decide to model this problem with fuzzy sets. How do you model the fuzzy set of "employees who live moderately far away", i.e. those who do not live close, but also not far away?

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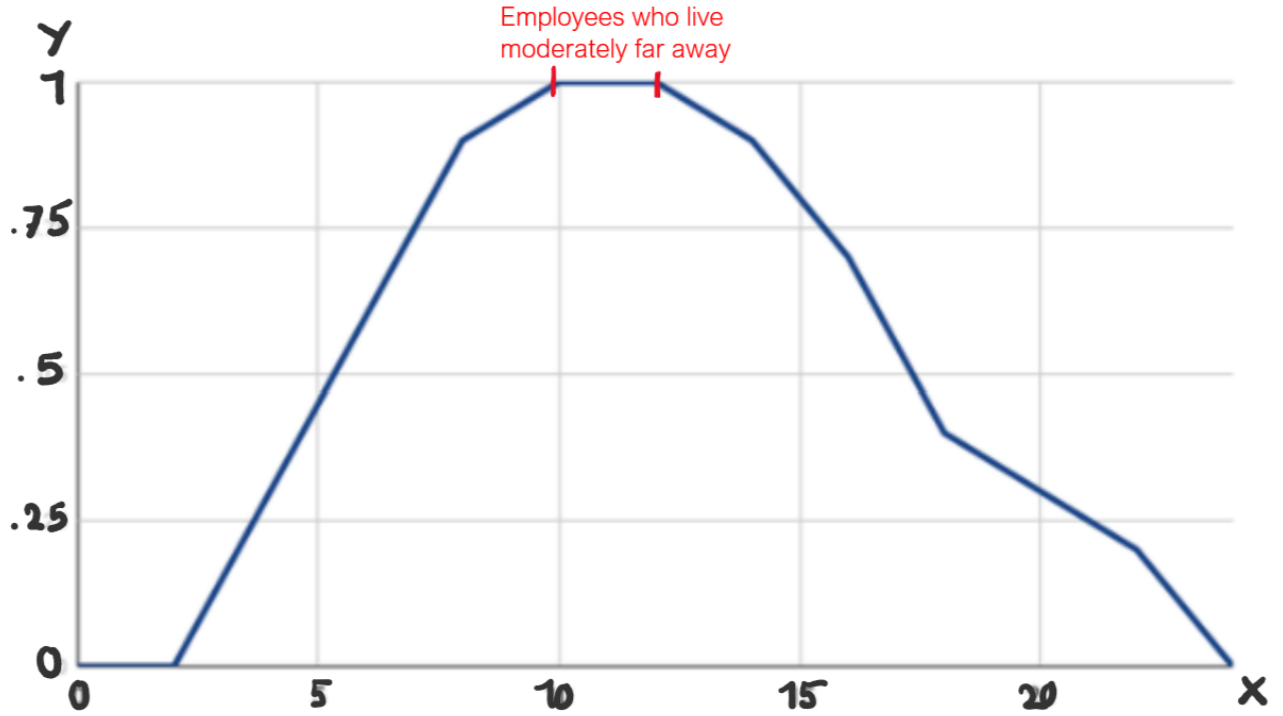
X – AXIS: Distance from workplace

Y – AXIS: Degree of membership (living moderately far away)

The degree of living moderately far away keeps increasing as the distance from the workplace increases.

At a certain point, this degree will drop since those employees live too far away to be considered only "**moderately far**".

Thus, using this model, the director could consider employees living at distances shown by the peak of the curve as "**moderately far**", since the membership is equal to 1 at that point.



Fuzzy Logic

Fuzzy Logic Operators

Variables have a degree of truth between 0 and 1, 0 meaning False and 1 meaning True.

The exercise provides the truth values for two variables, for example, P and Q or A and B.

Standard semantics of fuzzy logic:

- $f(\neg A) = 1 - f(A)$ Complement
- $f(A \vee B) = \max(f(A), f(B))$ Union
- $f(A \& B) = \min(f(A), f(B))$ Intersection
- $f(A \rightarrow B) = \min(1, 1 - f(A) + f(B))$ Implication

→ Alternative rules could possibly be applied, always check the semantics given in the exercise!

Fuzzy Logic Operators: Recap Question

Exercise 3: Given the following (standard) semantics for Fuzzy Logic for the following operators:

- f(¬A) = 1 - f(A)
- f(A v B) = max(f(A);f(B))
- f(A & B) = min(f(A);f(B))
- f(A -> B) = min(1;1 – f(A) + f(B))

What is the meaning of f((P v (P -> Q)) & (¬Q ->¬P)) if f(P)=0.8 and f(Q)=0.3?

Fuzzy Logic Operators: Recap Question

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What is the meaning of f((P v (P -> Q)) & (¬Q ->¬P)) if f(P)=0.8 and f(Q)=0.3?

The formula is: $f((P \vee (P \to Q))\&(\neg Q \to \neg P))$

1. **Implication ($P \to Q$):**
   $f(P \to Q) = \min(1, 1 - f(P) + f(Q))$
   $f(P \to Q) = \min(1, 1 - 0.8 + 0.3) = \min(1, 0.5) = 0.5$
2. **Disjunction ($P \vee (P \to Q)$):**
   $f(P \vee (P \to Q)) = \max(f(P), f(P \to Q))$
   $f(P \vee (P \to Q)) = \max(0.8, 0.5) = 0.8$
3. **Negation ($\neg Q$):**
   $f(\neg Q) = 1 - f(Q)$
   $f(\neg Q) = 1 - 0.3 = 0.7$
4. **Negation ($\neg P$):**
   $f(\neg P) = 1 - f(P)$
   $f(\neg P) = 1 - 0.8 = 0.2$
5. **Implication ($\neg Q \to \neg P$):**
   $f(\neg Q \to \neg P) = \min(1, 1 - f(\neg Q) + f(\neg P))$
   $f(\neg Q \to \neg P) = \min(1, 1 - 0.7 + 0.2) = \min(1, 0.5) = 0.5$
6. **Conjunction ($(P \vee (P \to Q))\&(\neg Q \to \neg P)$):**
   $f((P \vee (P \to Q))\&(\neg Q \to \neg P)) = \max(f(P \vee (P \to Q)), f(\neg Q \to \neg P))$
   $f((P \vee (P \to Q))\&(\neg Q \to \neg P)) = \max(0.8, 0.5) = 0.8$

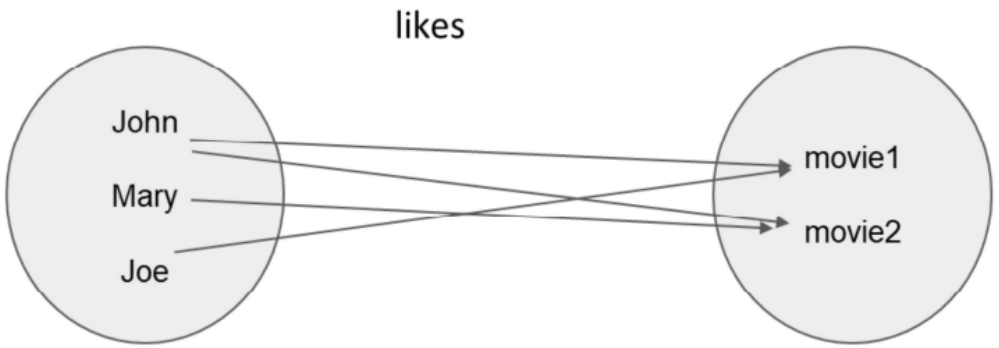
So, the correct meaning of $f((P \vee (P \to Q))\&(\neg Q \to \neg P))$ with $f(P) = 0.8$ and $f(Q) = 0.3$ is 0.8.

Fuzzy Relations

Fuzzy Relations

We go from a relation between pairs from two sets:

A = {John, Mary, Joe}
B = {movie1, movie2}



To a **fuzzy** relation between pairs from two sets:

A = {John, Mary, Joe}
B = {movie1, movie2}

likes = {(John, movie1), (John, movie2), (Mary, movie2), (Joe, movie1)}

How do we portray this?

Fuzzy Relations

Fuzzy relation matrix: Fuzzy relations are often represented using matrices. Each element of the matrix corresponds to the degree of membership between the elements of the sets.

A = {John, Mary, Joe}
B = {movie1, movie2}

likes = {(John, movie1), (John, movie2), (Mary, movie2), (Joe, movie1)}

likes =

	movie1	movie2
John	1	1
Mary	0	1
Joe	1	0

Fuzzy Relations

Fuzzy composition: $R \circ S$ or R ; S stands for the composition of R and S.

Given the two fuzzy relations R and S over sets x,y and y,z respectively, the composition $R \circ S$ is a fuzzy relation from x to z which is defined by the following:

$$\boxed{f_{R \circ S}(x, z) = \max_y \left(\min(f_R(x, y), f_S(y, z))\right)}$$

Fuzzy Relations: Recap Question

Exercise 4: Suppose that there are two relations defined as follows, where L5 and L6 stand for Lecture 5 and 6 respectively:

understands

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
Julia	0.1	0.4

contains

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

Fill in the table for the composition of the two relations, i.e. the relation

Understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

Fuzzy Relations: Recap Question

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L5	0.8	0.3
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Fill in the table for the composition of the two relations, i.e. the relation

Understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

The two **sets** in this question are:

- The people: (John, Peter, Julia)
- The lectures: (L5,L6)

→ lecture 5 and lecture 6

The two **relations** are:

- *understands:* indicates to what degree the person understands the lecture
- *contains:* indicates how much semantic and reasoning content each lecture contains

Fuzzy Relations: Recap Question

Exercise 4: Suppose that there are two relations defined as follows, where L5 and L6 stand for Lecture 5 and 6 respectively:

understands

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
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contains

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

Fill in the table for the composition of the two relations, i.e. the relation

Understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

Now we need to find the value of the **composed relation** for the pair (x,z) using:

$$f_{R \circ S}(x, z) = \max_y \left(\min(f_R(x, y), f_S(y, z))\right)$$

Where *x* is a person and *z* is semantics or reasoning. We consider all possible lectures, represented by *y*.

Calculate all **composed relation values**:

- John, semantics
- John, reasoning
- Peter, semantics
- etc.

Fuzzy Relations: Recap Question

$$f_{R\circ S}(x,z) = \max_y \left(\min(f_R(x,y), f_S(y,z))\right)$$

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
Julia	0.1	0.4

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

John, semantics = max(min(**0.4**,**0.8**),min(**0.6**,**0.4**) = **0.4**

John, reasoning = max(min(0.4,0.3),min(0.6,0.8) = 0.6

Peter, semantics = max(min(0.3,0.8),(0.7,0.4) = 0.4

Peter, reasoning = max(min(0.3,0.3),(0.7,0.8)) = 0.7

Julia, semantics = max(min(0.1,0.8),min(0.4,0.4)) =0.4

Julia, reasoning = max(min(0.1,0.3),min(0.4,0.8) = 0.4

Final Answer:

	semantics	reasoning
John	0.4	0.6
Peter	0.4	0.7
Julia	0.4	0.4

Exam Questions

Exam Question 1: Uncertainty and Vagueness

Consider the following statement: “All Dutch men are tall”. Which of the following statements is true?

A) The author of this statement describes the height of Dutch men with a degree of uncertainty.

B) The author of this statement describes the height of Dutch men with a degree of vagueness.

Exam Question 1: Uncertainty and Vagueness

Consider the following statement: “All Dutch men are tall”. Which of the following statements is true?

A) The author of this statement describes the height of Dutch men with a degree of uncertainty.

B) The author of this statement describes the height of Dutch men with a degree of vagueness.

The statement "All Dutch men are tall" suggests a generalization without providing specific details or a clear definition of what is considered "tall." The term "tall" is subjective and lacks precision in this context, indicating vagueness rather than uncertainty.

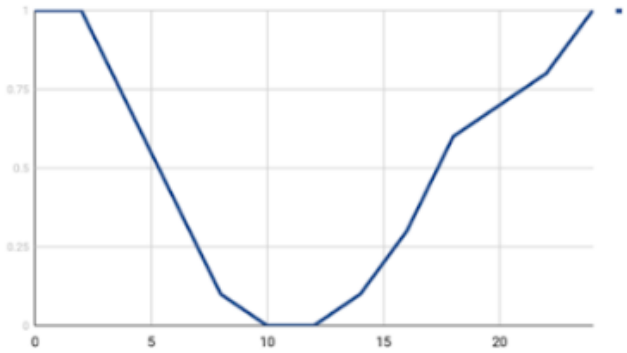
Exam Question 2: Fuzzy Sets

The director of our institute wants to reduce the environmental impact of the company by reducing the CO2 emissions caused by transportation. Therefore, she wants to give a bonus to employees driving a very clean car. however, she does not want to pay those who live close (they can come by bicycle) or those who live very far (they should do remote work).

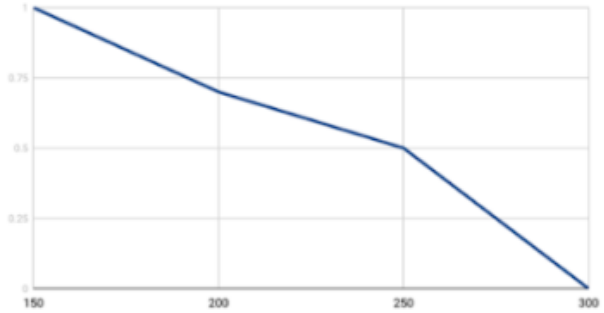
Since you know the director is a big fan of fuzzy set theory, you decide to solve this problem with fuzzy sets.

How would you model the fuzzy set of clean cars (what could the membership function look like? What is on the x-axis?) You can freely choose the shape of the curve, but it needs to be such that cleaner cars have a stronger membership compared to more polluting cars. Which of the courses is appropriate for this problem?

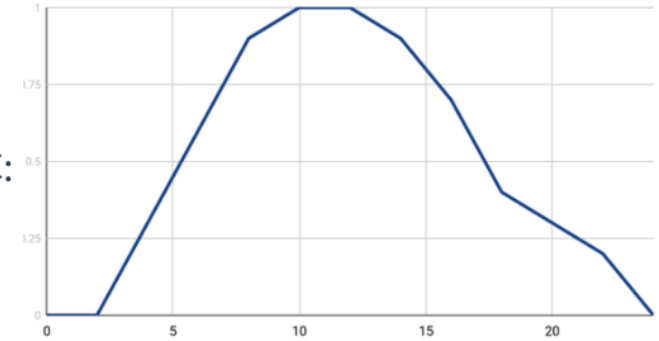
A:



B:



C:



Exam Question 2: Fuzzy Sets

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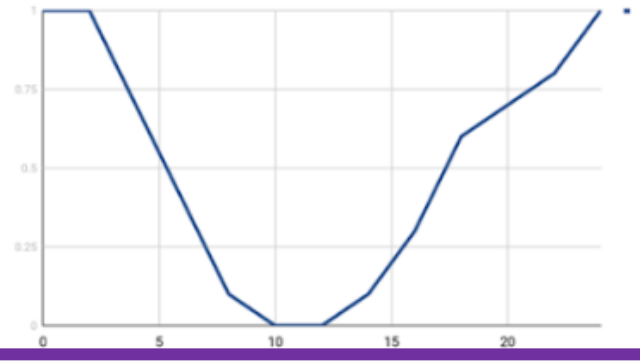
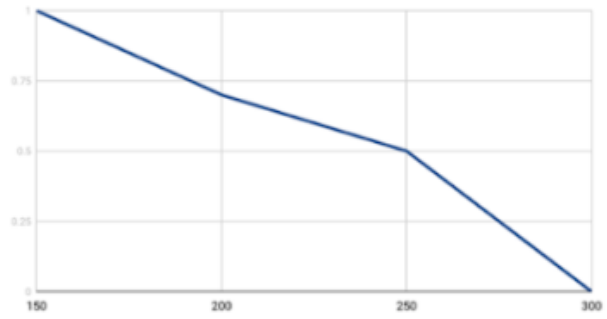


B:



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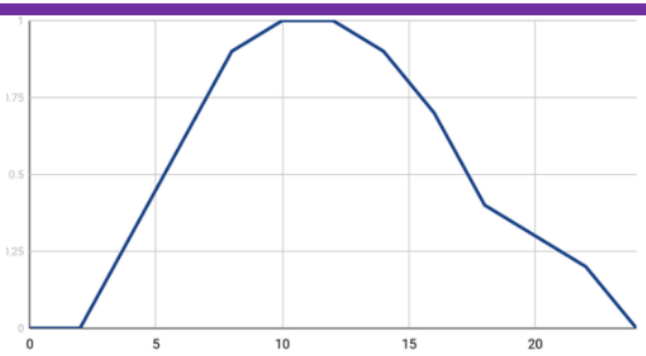

Exam Question 3: Fuzzy Semantics

Given the following (standard) semantics for Fuzzy Logic for the following operators:

- $f(\neg A) = 1 - f(A)$
- $f(A \vee B) = \max(f(A), f(B))$
- $f(A \& B) = \min(f(A), f(B))$
- $f(A \rightarrow B) = \min(1, 1 - f(A) + f(B))$

What would be the fuzzy interpretation of the formula

$$\neg A \vee (A \vee \neg B)) \rightarrow (B \rightarrow \neg A))$$

given that $f(A)=0.4$ and $f(B)=0.6$.

Write down the necessary steps and the result.

Exam Question 3: Fuzzy Semantics

Given the following (standard) semantics for Fuzzy Logic for the following operators:

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- f(A -> B) = min(1,1 - f(A) + f(B))

What would be the fuzzy interpretation of the formula

¬ A v (A v ¬ B)) -> (B ->¬A))

given that f(A)=0.4 and f(B)=0.6.

Write down the necessary steps and the result.

The formula is: $\neg A \vee (A \vee \neg B) \rightarrow (B \rightarrow \neg A)$

1. **Negation ($\neg A$):**
   $f(\neg A) = 1 - f(A)$
   $f(\neg A) = 1 - 0.4 = 0.6$
2. **Disjunction ($A \vee \neg B$):**
   $f(A \vee \neg B) = \max(f(A), f(\neg B))$
   $f(\neg B) = 1 - f(B) = 1 - 0.6 = 0.4$
   $f(A \vee \neg B) = \max(0.4, 0.6) = 0.6$
3. **Disjunction ($\neg A \vee (A \vee \neg B)$):**
   $f(\neg A \vee (A \vee \neg B)) = \max(f(\neg A), f(A \vee \neg B))$
   $f(\neg A \vee (A \vee \neg B)) = \max(0.6, 0.6) = 0.6$
4. **Implication ($(B \rightarrow \neg A)$):**
   $f(B \rightarrow \neg A) = \min(1, 1 - f(B) + f(\neg A))$
   $f(B \rightarrow \neg A) = \min(1, 1 - 0.6 + 0.6) = \min(1, 1) = 1$
5. **Implication ($\neg A \vee (A \vee \neg B)) \rightarrow (B \rightarrow \neg A)$:**
   $f(\neg A \vee (A \vee \neg B) \rightarrow (B \rightarrow \neg A)) = \min(1, 1 - f(\neg A \vee (A \vee \neg B)) + f(B \rightarrow \neg A))$
   $f(\neg A \vee (A \vee \neg B) \rightarrow (B \rightarrow \neg A)) = \min(1, 1 - 0.6 + 1) = \min(1, 1.4) = 1$

So, the fuzzy interpretation of the given formula $\neg A \vee (A \vee \neg B) \rightarrow (B \rightarrow \neg A)$ with $f(A) = 0.4$ and $f(B) = 0.6$ is 1.

Exam Question 4: Fuzzy Set Theory

Unlike normal sets, fuzzy sets cannot be combined by union, true or false?

A) True

B) False

Exam Question 4: Fuzzy Set Theory

Unlike normal sets, fuzzy sets cannot be combined by union, true or false?

A) True

B) False

Fuzzy sets allow for degrees of membership, where elements can belong to a set with varying degrees of membership between 0 and 1. The union operation in fuzzy set theory involves combining the degree of membership of elements from different fuzzy sets.

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I hope this helps ☺ You can send me any questions in the WhatsApp group if needed

Good luck with studying for the exam!